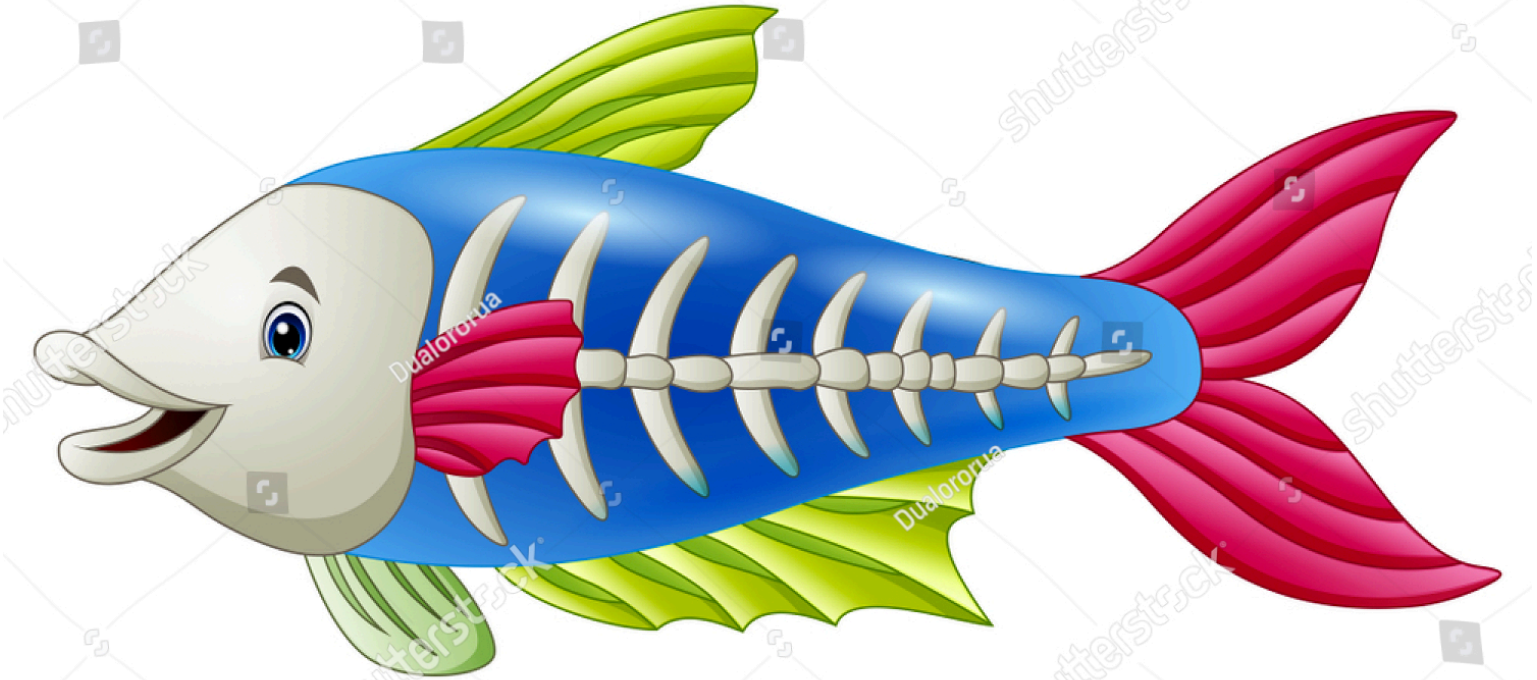
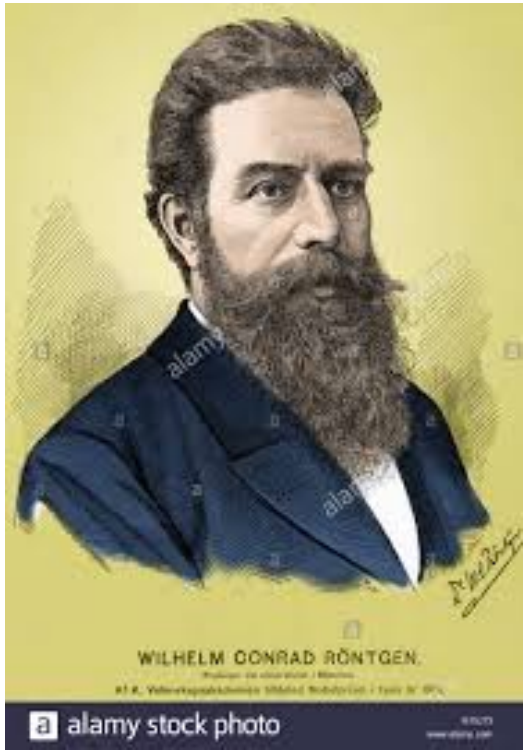
Atomic Physics

Chapter 6 X ray

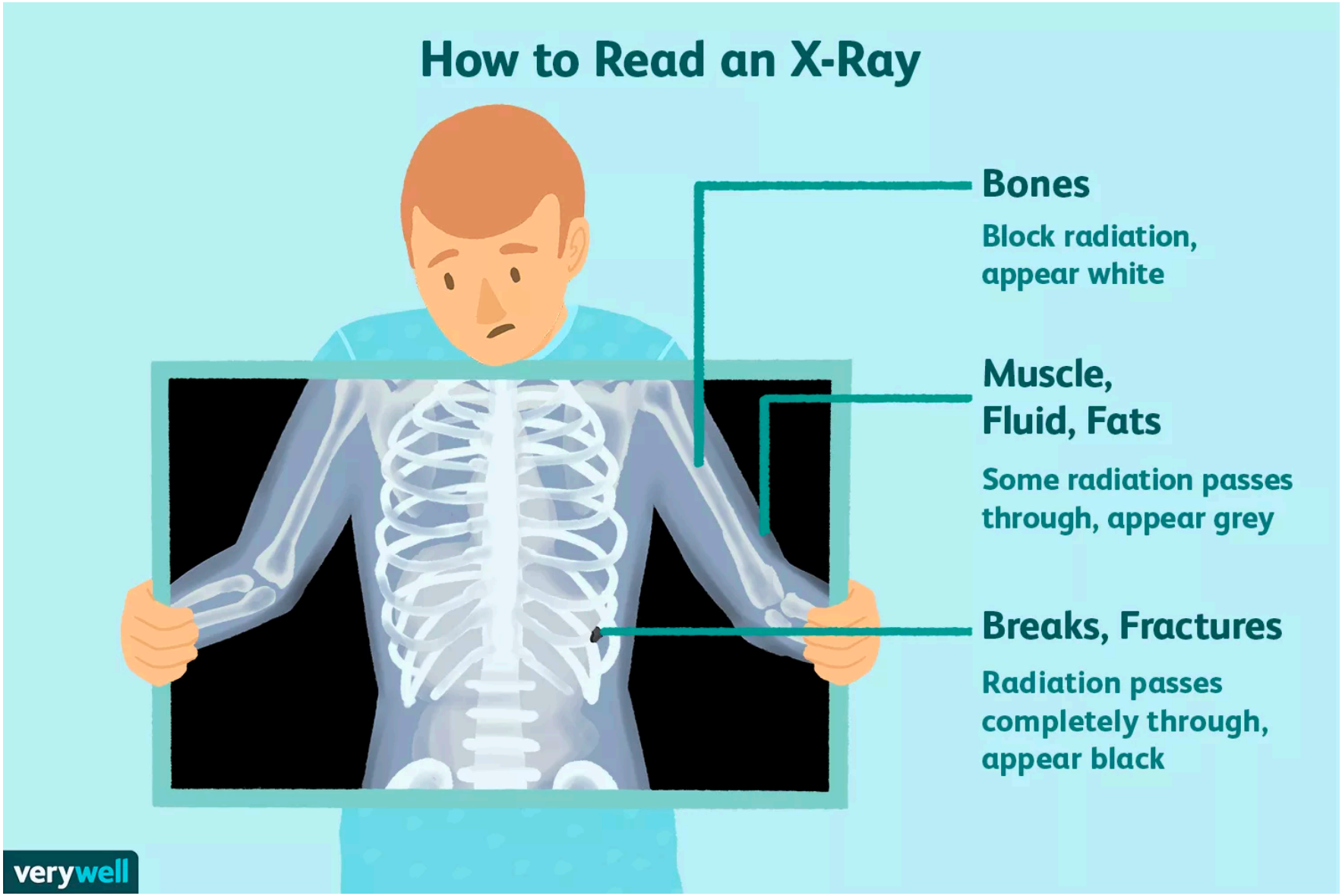


6.1 The discovery of X ray

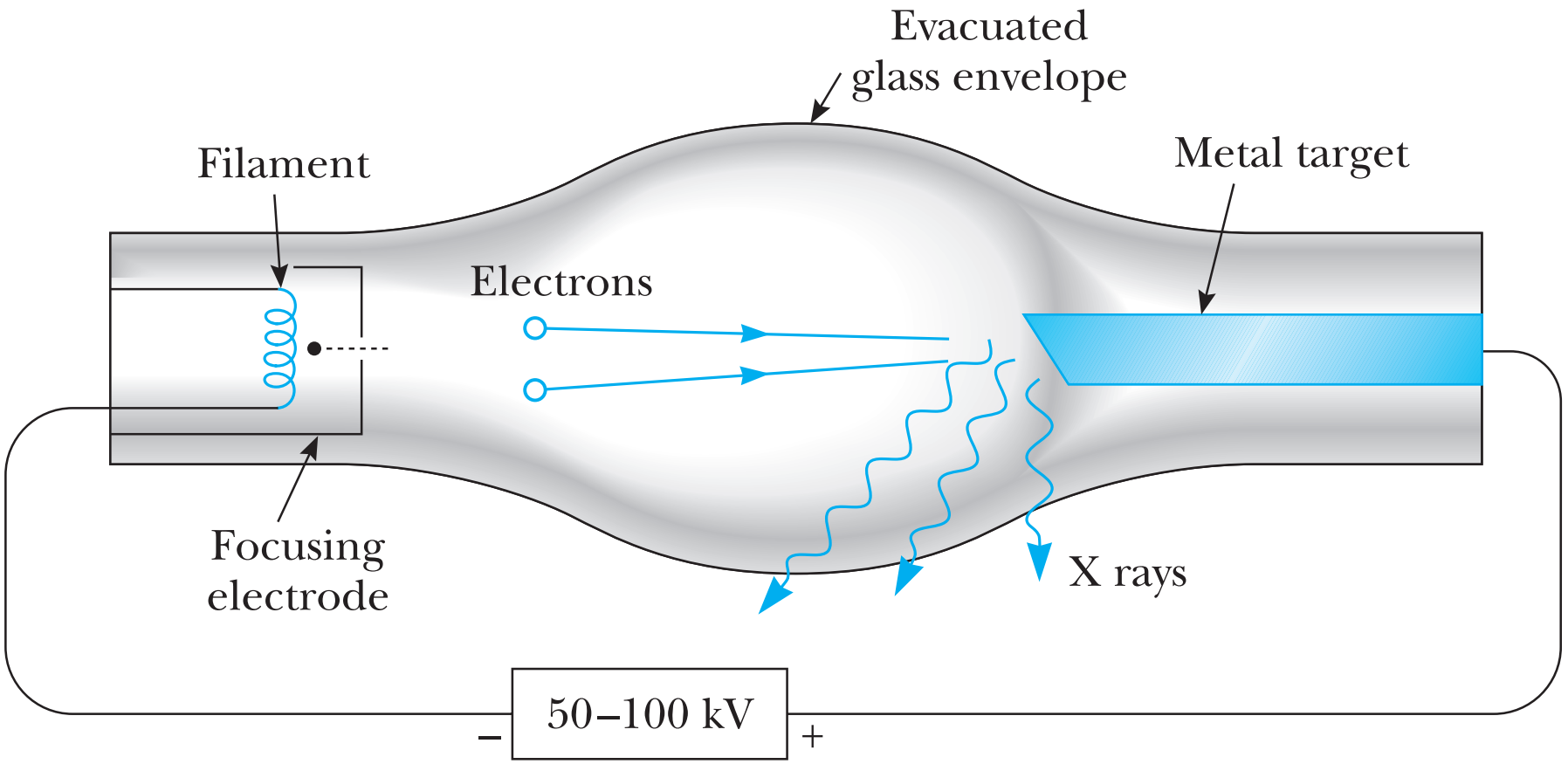
X-rays were discovered in 1895 by the German physicist Wilhelm Roentgen. He found that a beam of high-speed electrons striking a metal target produced a new and extremely penetrating type of radiation.





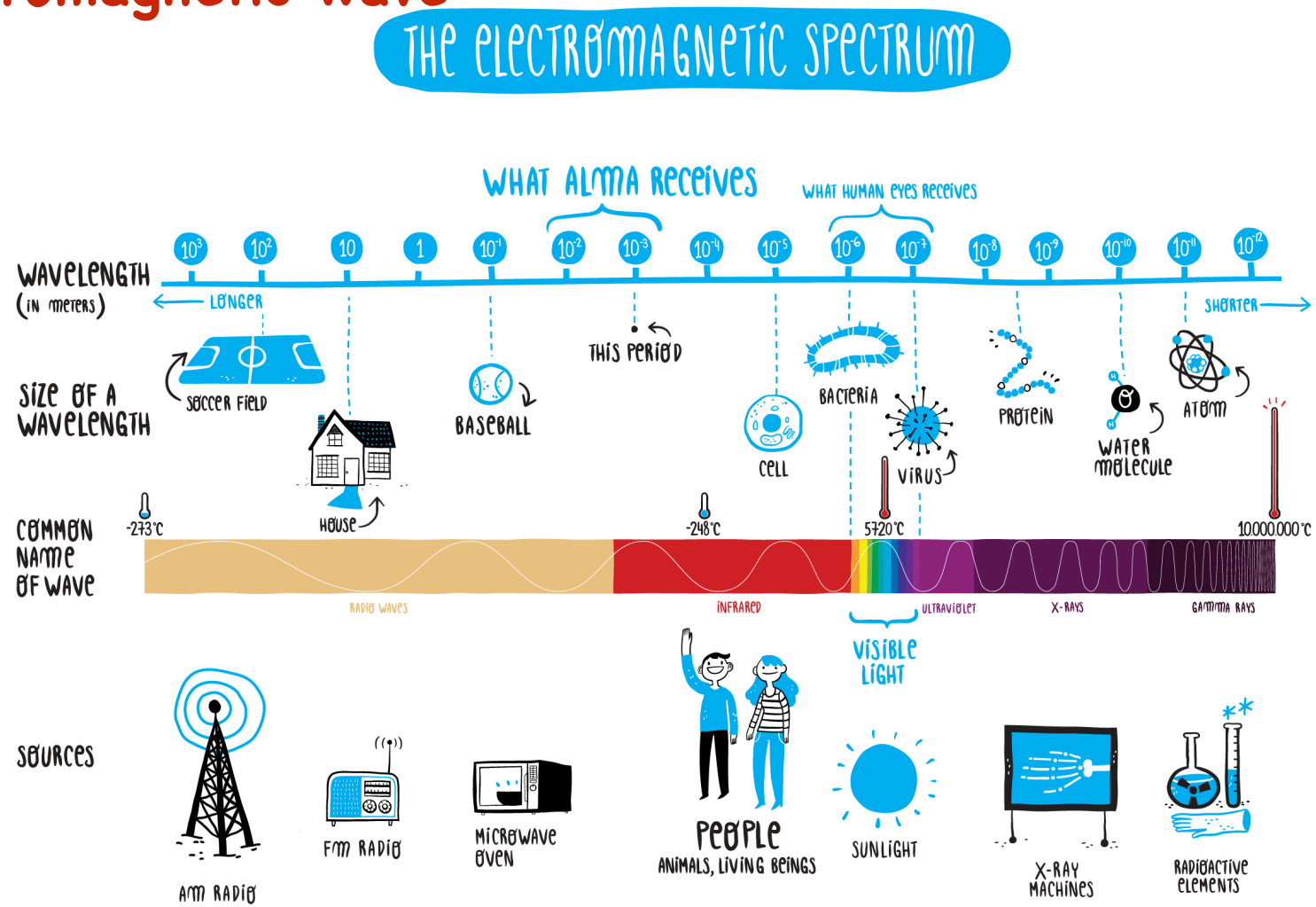

How to Read an X-Ray
Bones
Block radiation, appear white
Muscle, Fluid, Fats
Some radiation passes through, appear grey
Breaks, Fractures
Radiation passes completely through, appear black
verywell

X-ray Tube

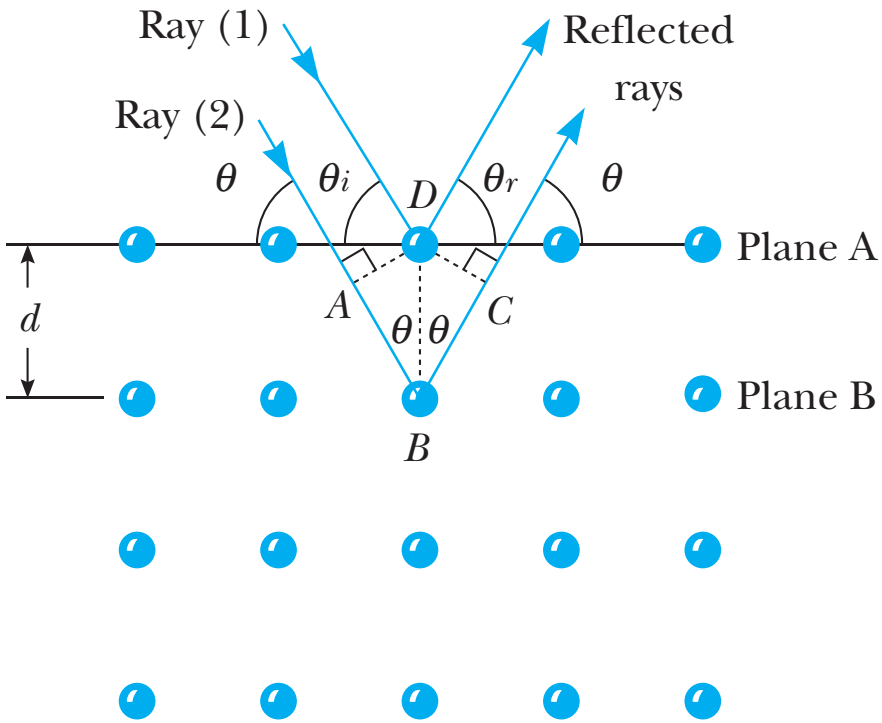

Evacuated
glass envelope
Filament
Metal target
Electrons
Focusing
electrode
X rays
50–100 kV
–
+

6.1 The discovery of X ray

Electromagnetic wave

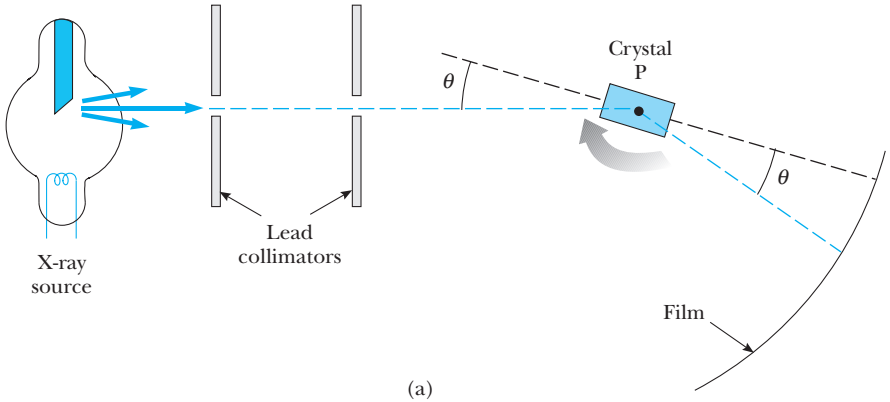


X ray diffraction

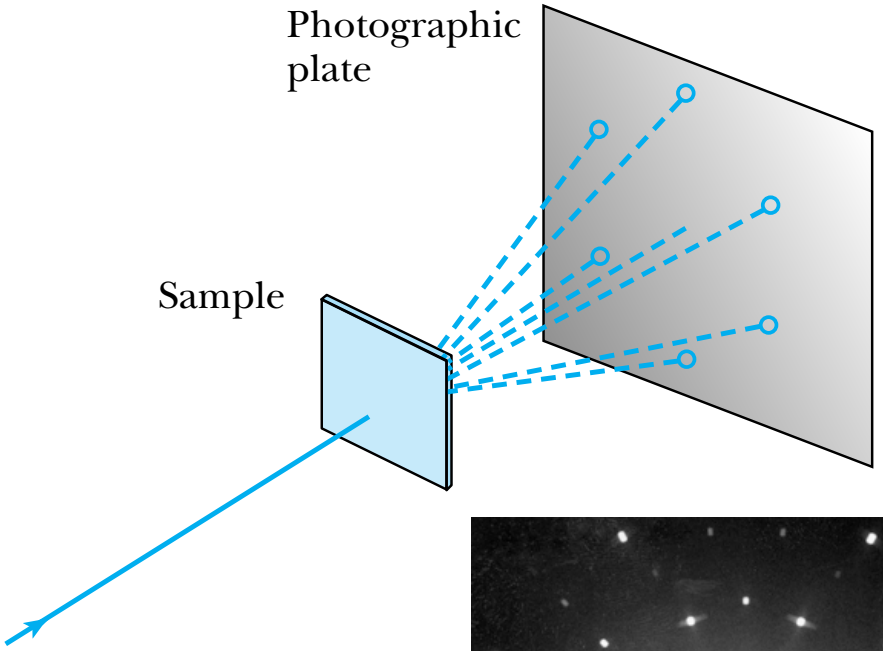


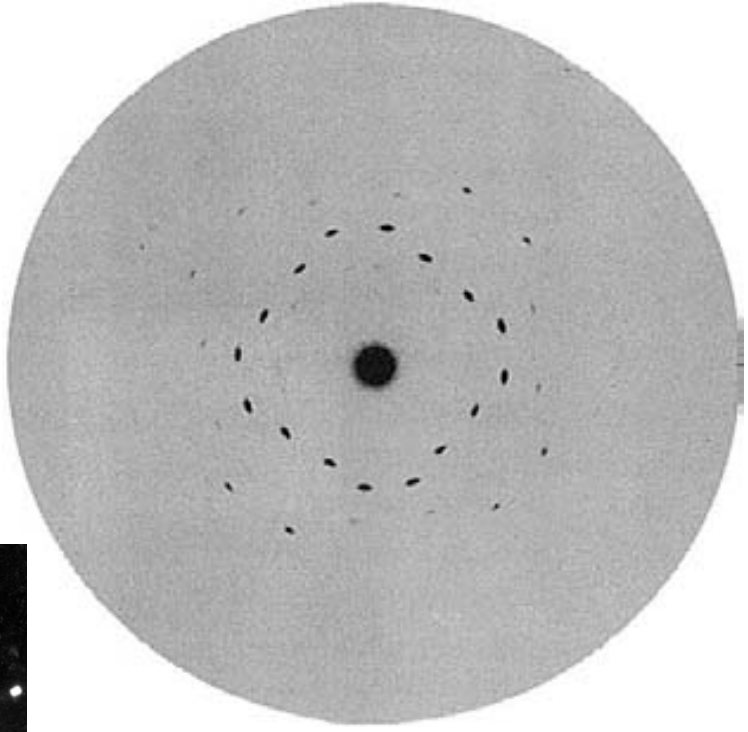
Bragg equation

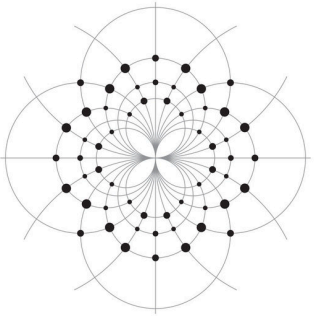
$$n\lambda = 2d \sin\theta \qquad n = 1, 2, 3, \ldots$$

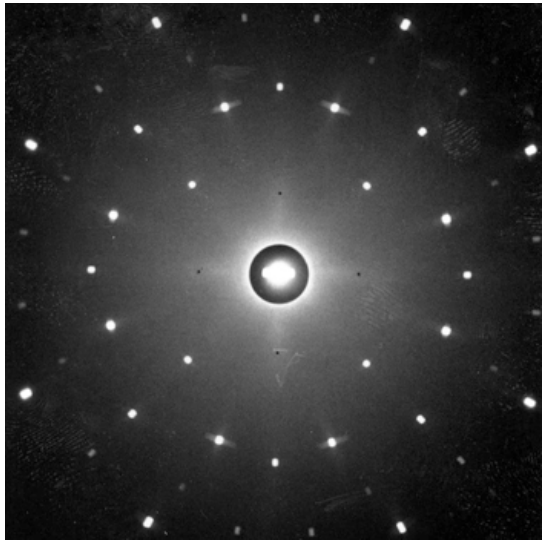


Laue diffraction transmission method



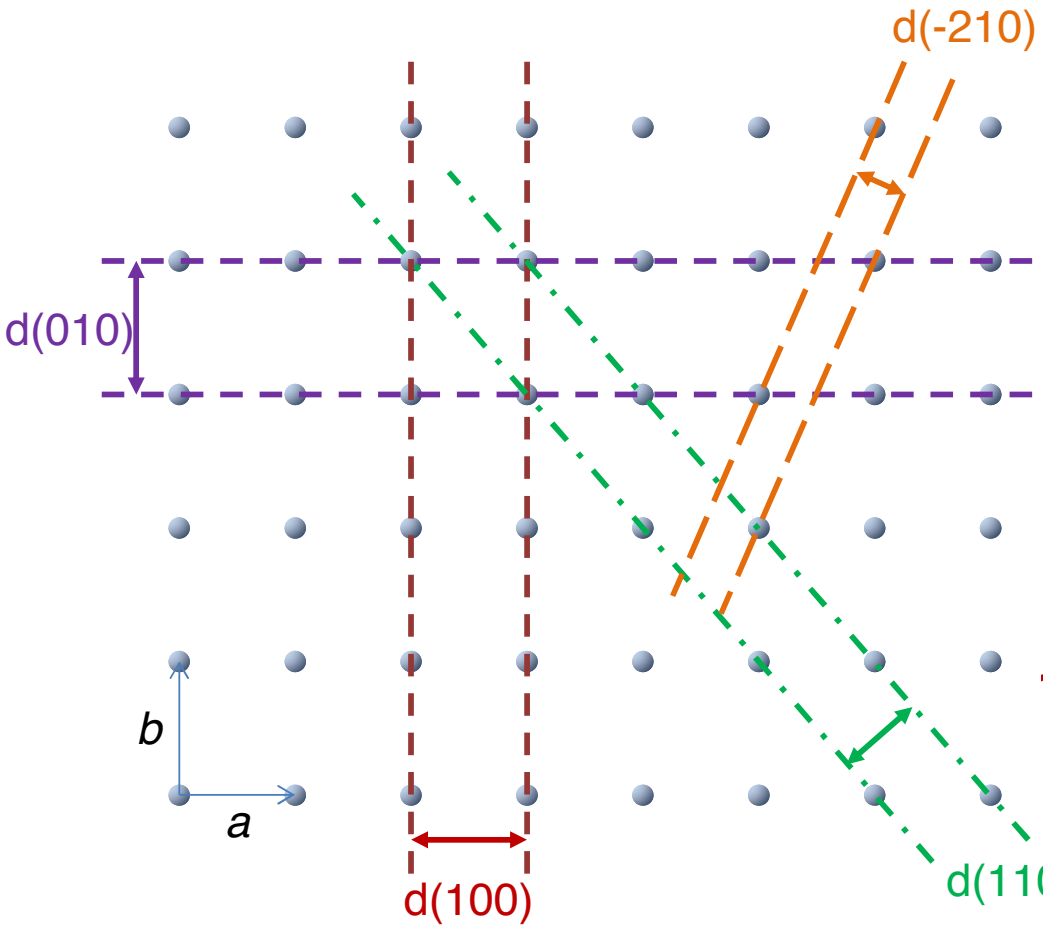


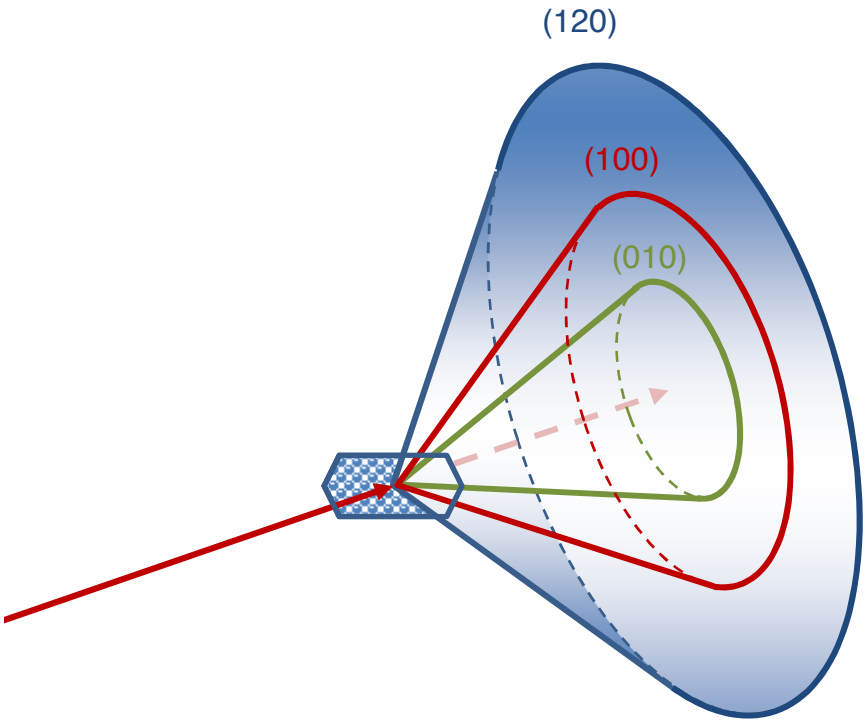




The diffraction of X ray

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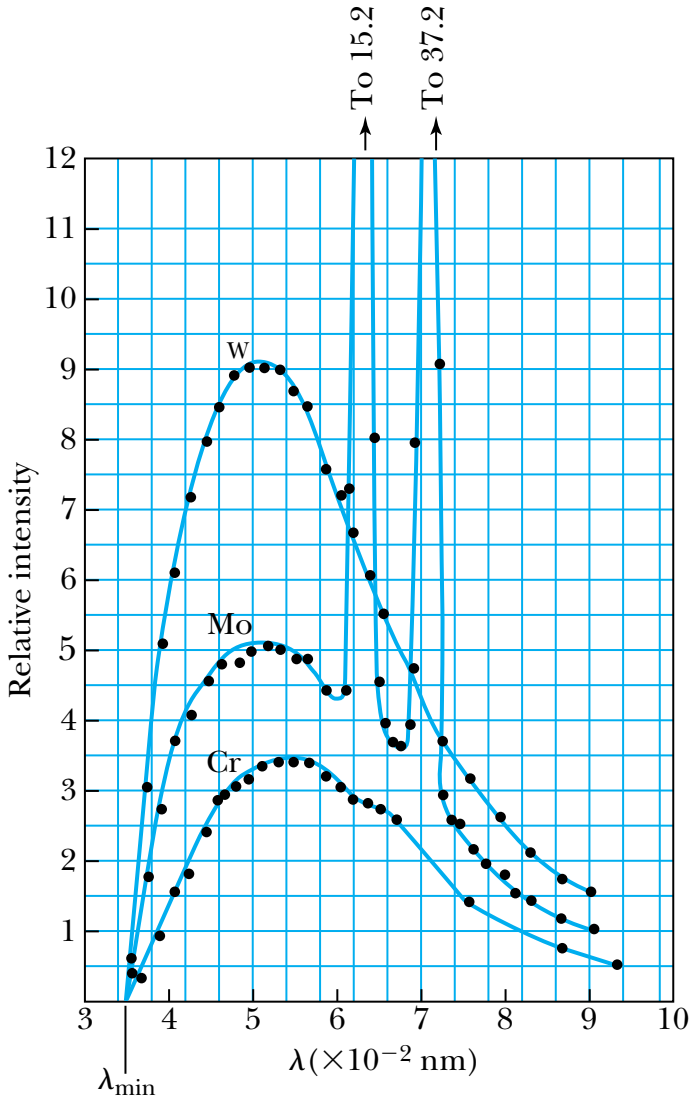
6.1 The discovery of X ray

The polarization of X ray

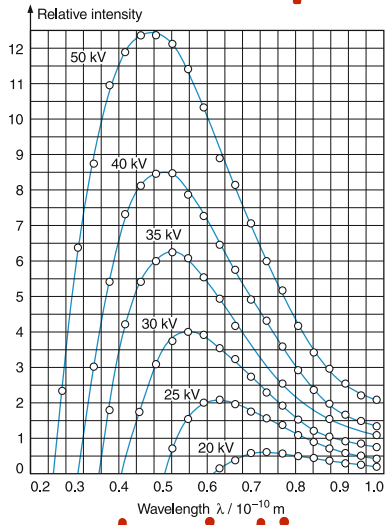


6.2 X ray production

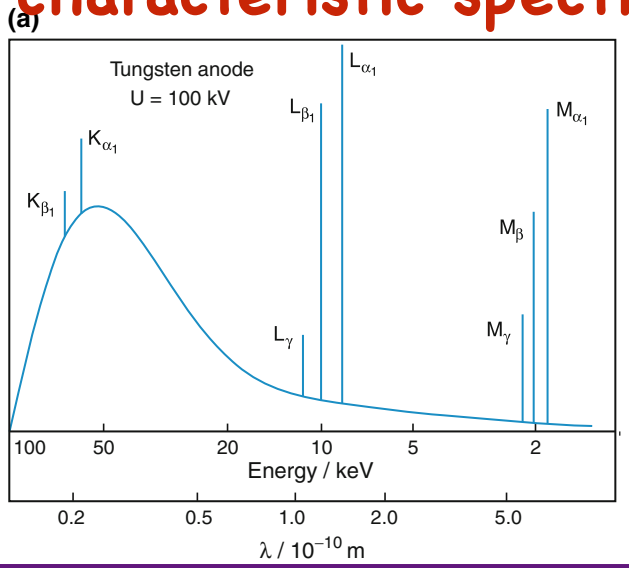
The spectrum of X ray



The continuous spectrum



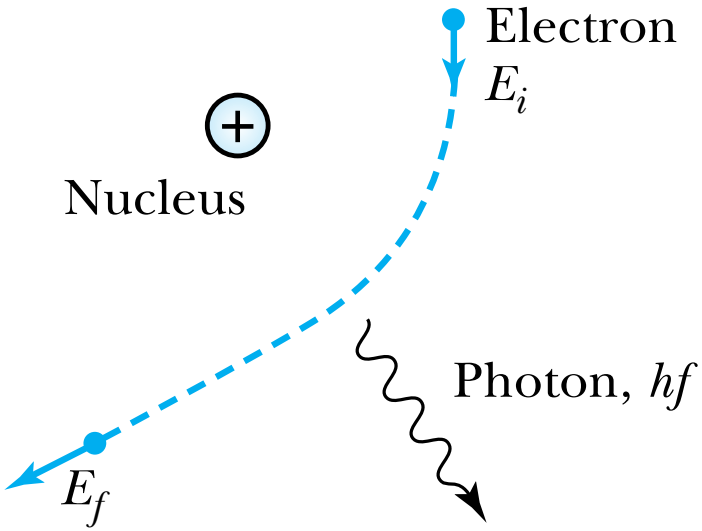
The characteristic spectrum

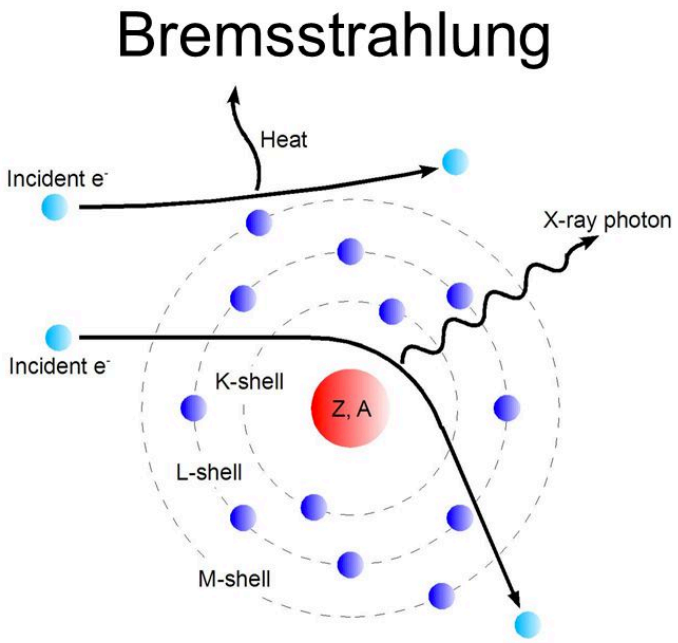


6.2 X ray production

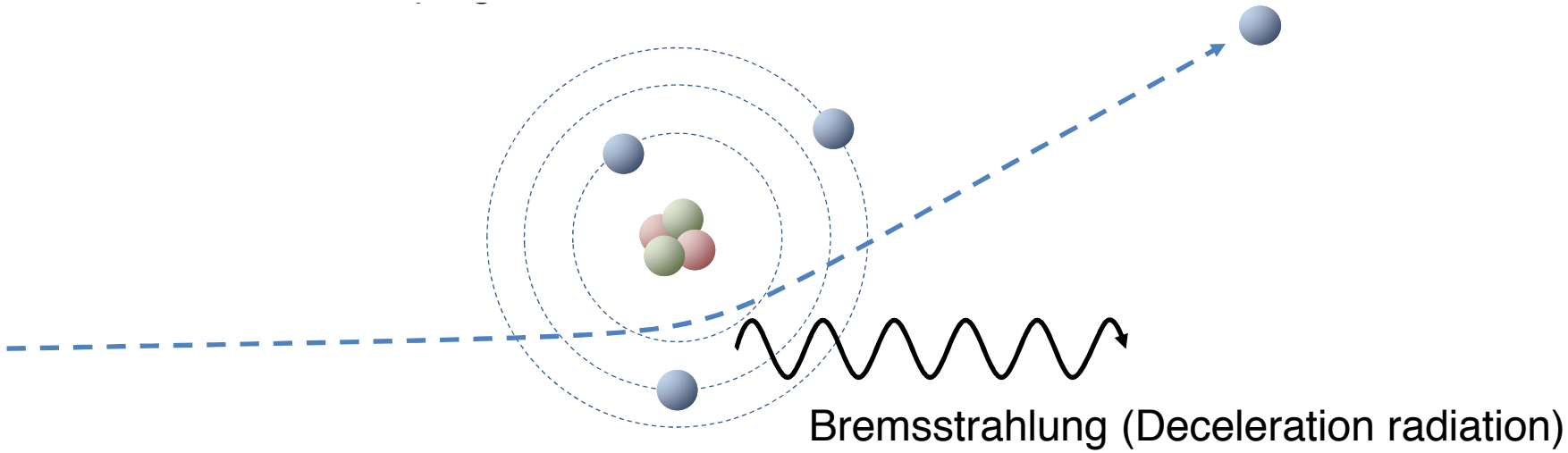
The continuous spectrum:

An energetic electron passing through matter will radiate photons and lose kinetic energy. The process by which photons are emitted by an electron slowing down is called bremsstrahlung, from the German word for "braking radiation."





The continuous spectrum



Electron is deflected and decelerated by the atomic nucleus. (Inelastic scattering).
Deflected electron emits electromagnetic radiation. Wavelength depends on the loss of energy.

The minimum wavelength is due to the inverse photoelectric effect. The conservation of energy requires that the electron kinetic energy equal the maximum photon energy:

$$eV_0 = hf_{\max} = \frac{hc}{\lambda_{\min}}$$

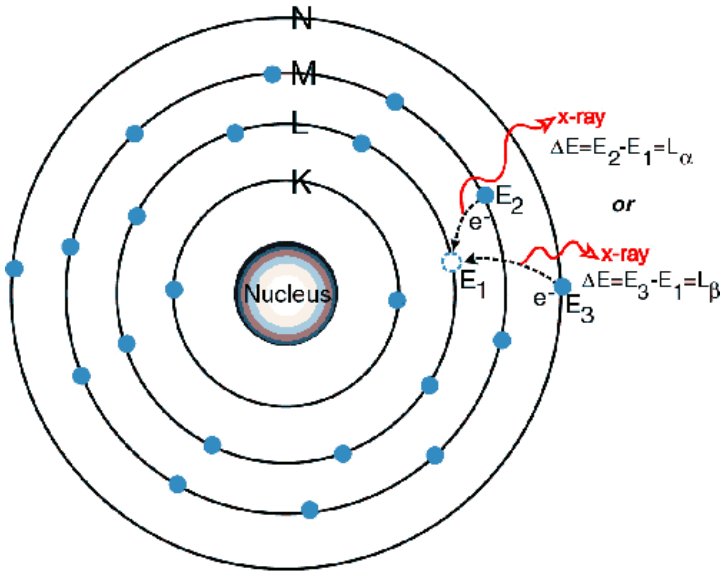
Therefore, the minimum wavelength (Duane-Hunt rule) is

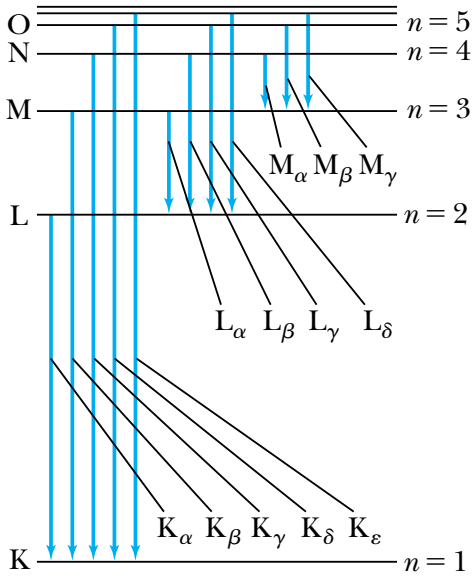
$$\lambda_{\min} = \frac{hc}{e}\frac{1}{V_0} = \frac{1.240 \times 10^{-6}\,\mathrm{V \cdot m}}{V_0}$$

6.2 X ray production

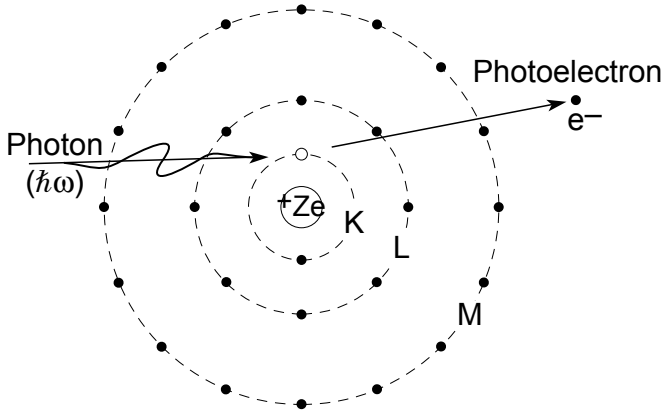
The characteristic spectrum:

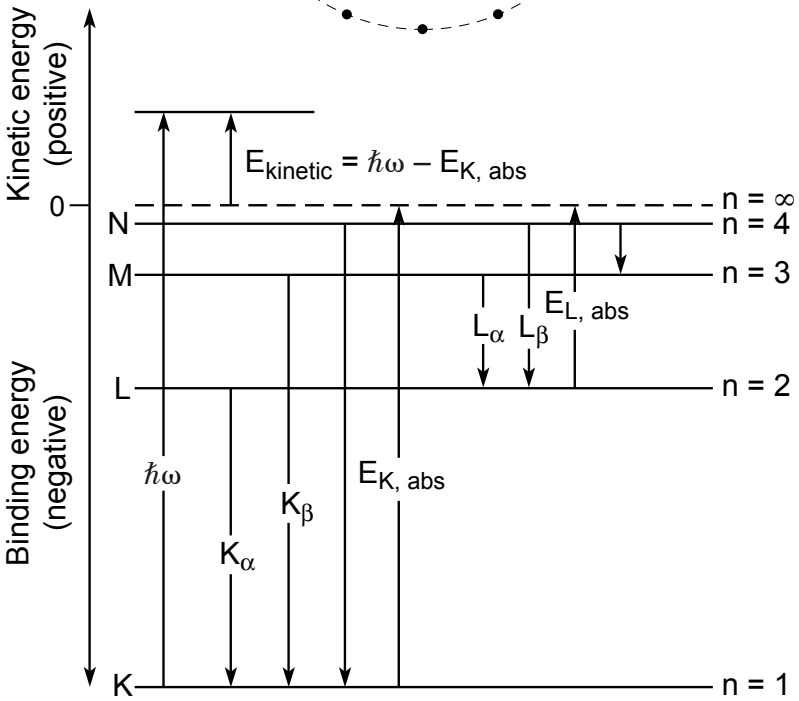
The atom is most stable in its lowest energy state or ground state, so it is likely that an electron from one of the higher shells will change its state and fill the inner-shell vacancy at lower energy, emitting radiation as it changes its state.

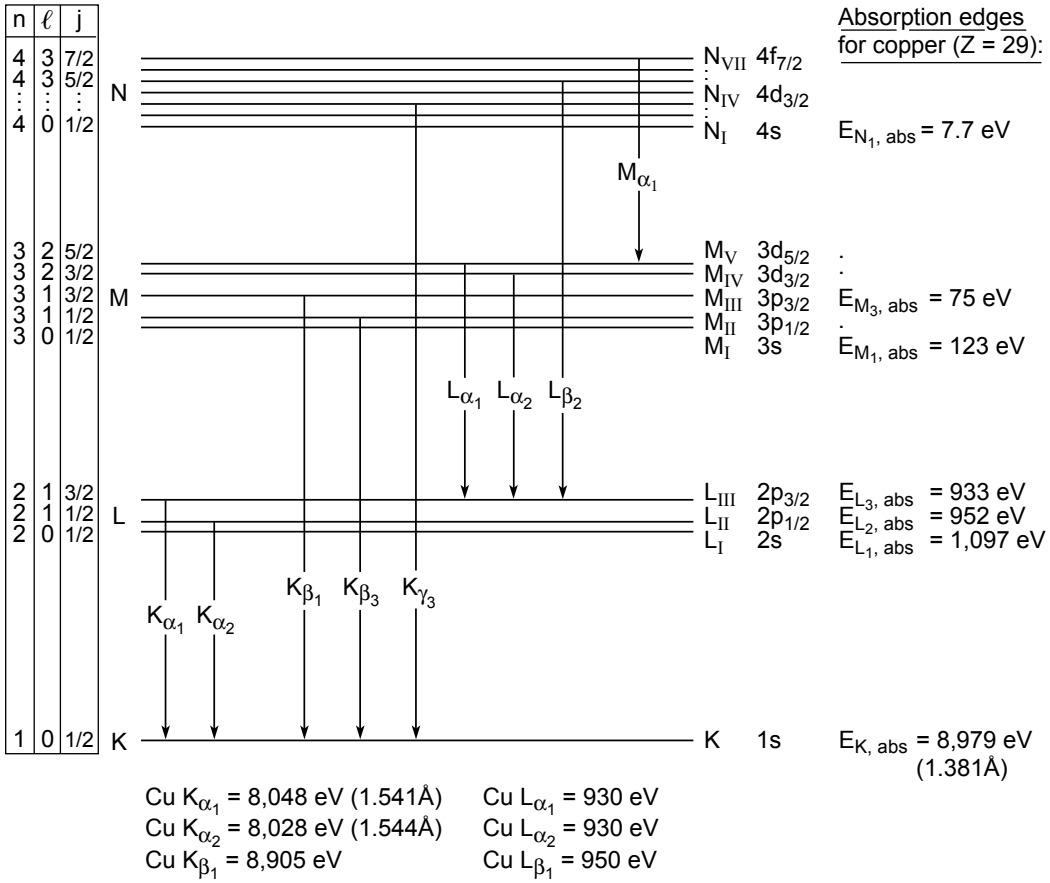




The characteristic spectrum







6.2 X ray production

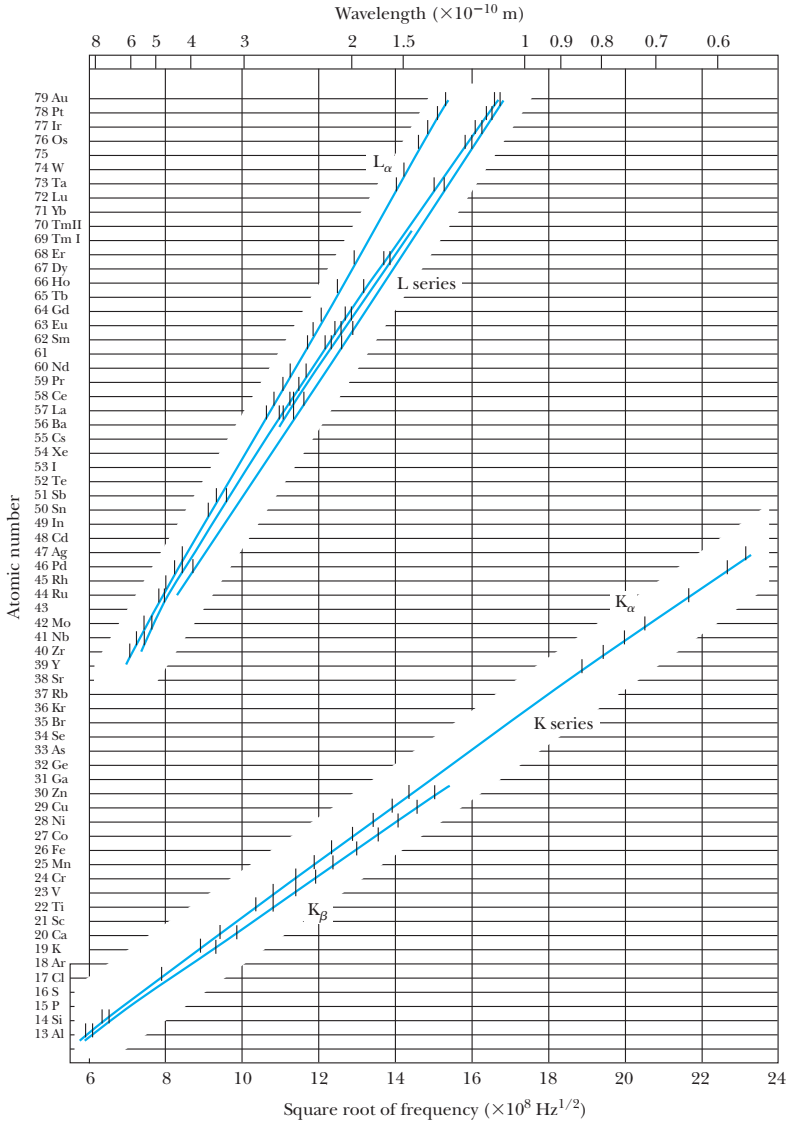
Moseley formula

$$\frac{1}{\lambda_{K_\alpha}} = R(Z-1)^2\left(\frac{1}{1^2}-\frac{1}{2^2}\right) = \frac{3}{4}R(Z-1)^2$$

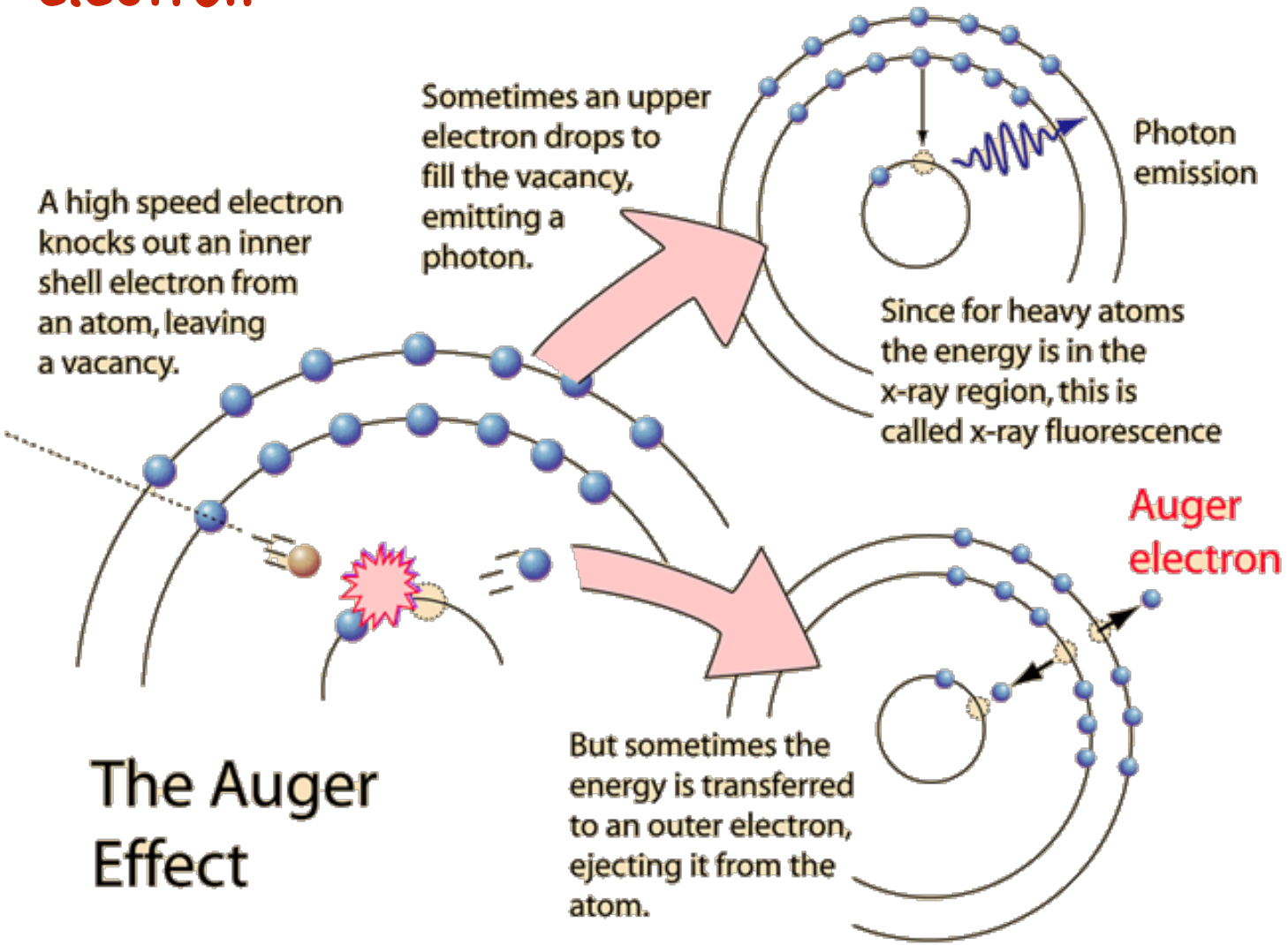
$$f_{K_\alpha} = \frac{c}{\lambda_{K_\alpha}} = \frac{3cR}{4}(Z-1)^2$$

$$\frac{1}{\lambda_K} = R(Z-1)^2\left(\frac{1}{1^2}-\frac{1}{n^2}\right) = R(Z-1)^2\left(1-\frac{1}{n^2}\right)$$

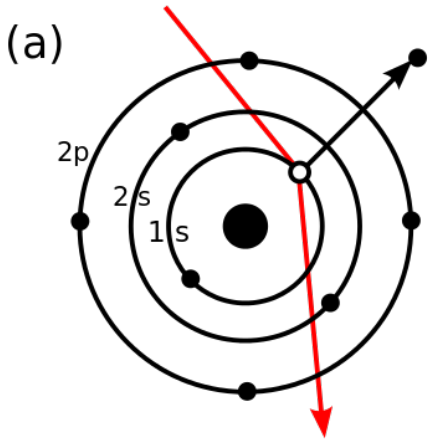




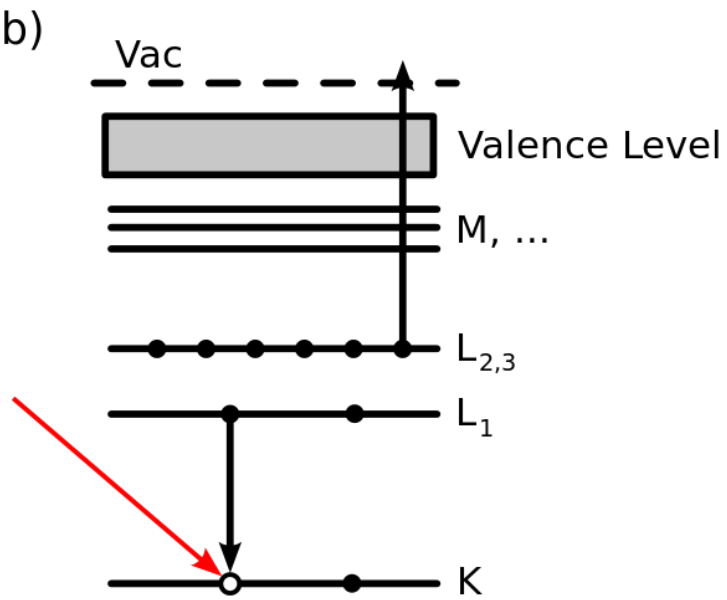
Auger electron

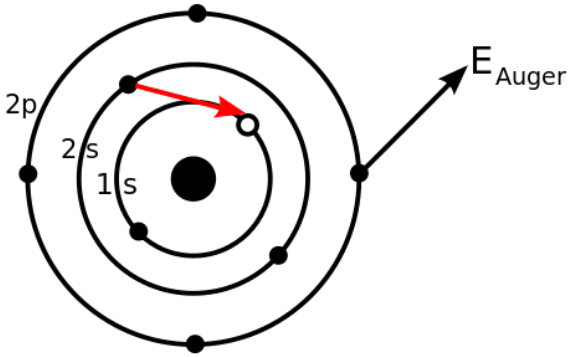


Auger electron



Electron collision





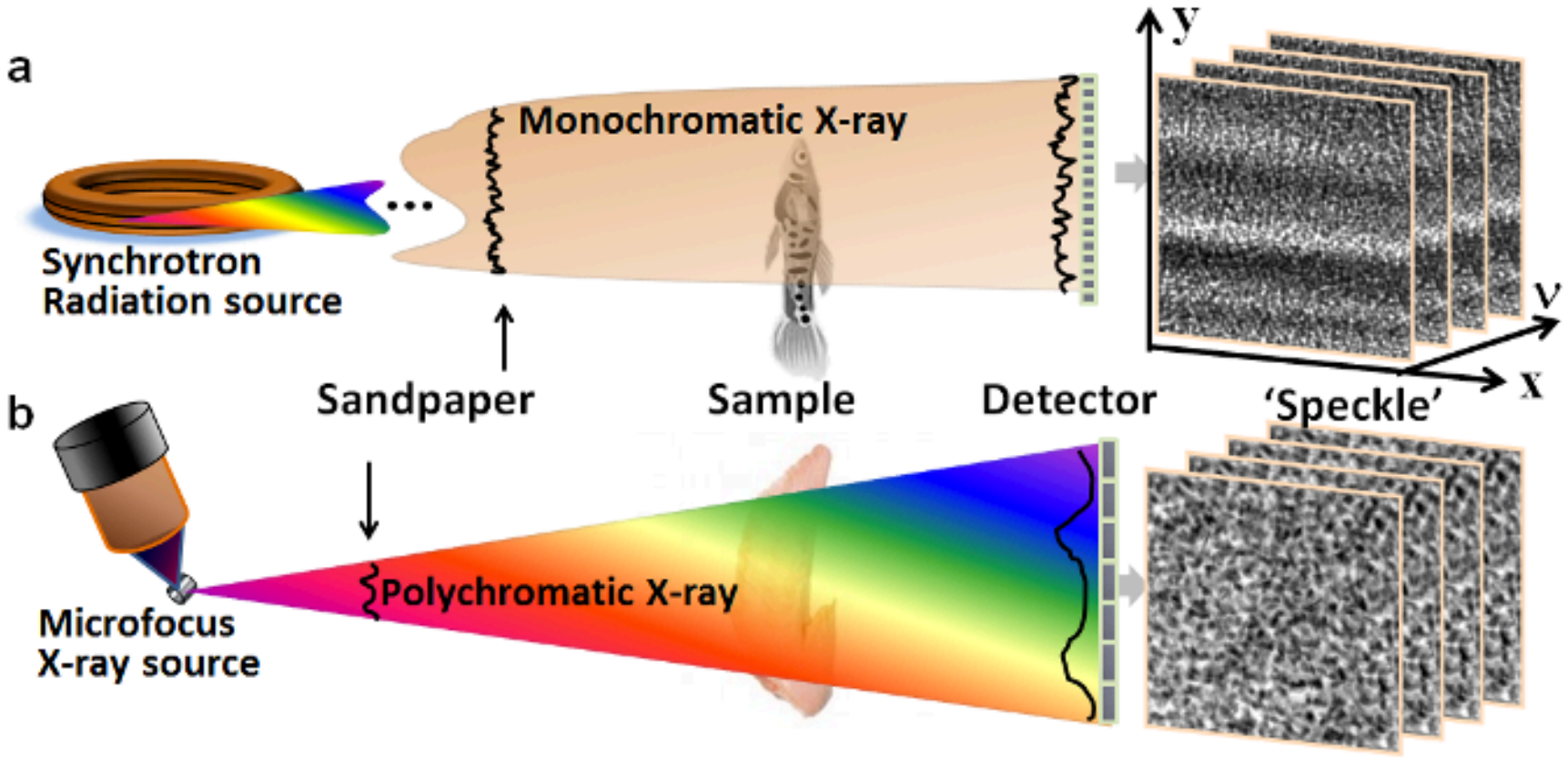
Auger electron emission

The kinetic energy of Auger electron

$$E_{ae} = E_K - E_{L_1} - E_{L_{2,3}}$$

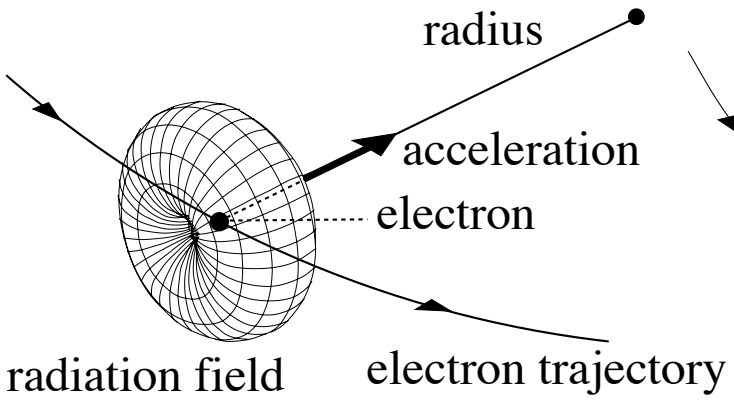
6.2 X ray production

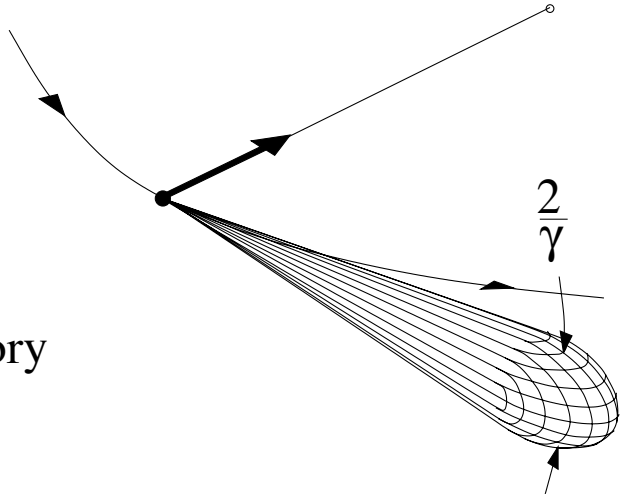
Synchrotron Radiation: In cyclic accelerators, when charged particles are accelerated, they radiate electromagnetic energy called synchrotron radiation.



Synchrotron Radiation

Rest frame → Lorentz transformation → Laboratory frame





$$\gamma = \frac{1}{\sqrt{1-\left(\frac{v}{c}\right)^2}} = \frac{E}{m_0 c^2}$$

Opening angle

Synchrotron Radiation

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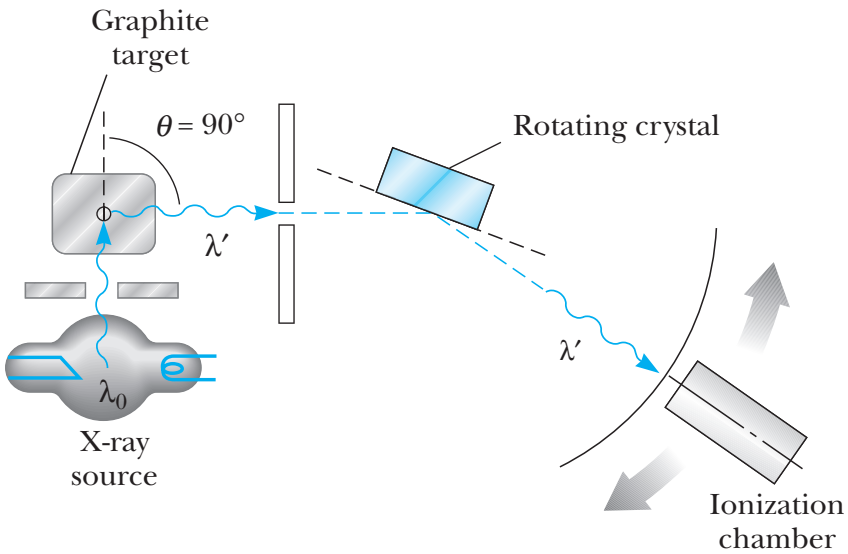
The very high intensity of the source yields images with a high signal-to-noise ratio on short time-scales, which enables fast radiographic investigations.

The beam can be easily monochromated. This allows correlations between attenuation values and the chemical constituents of the sample

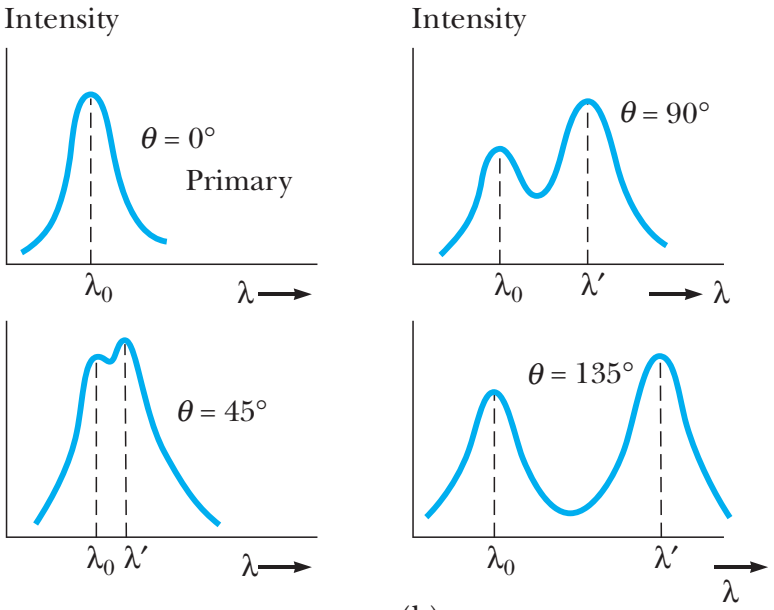
The option to vary the energy of the radiation enables the investigation of objects with very different absorption coefficients within the same measuring environment.

6.3 Compton scattering

At backward-scattering angles, there appeared to be a component of the emitted radiation (called a modified wave) that had a longer wavelength than the original primary (unmodified) wave.

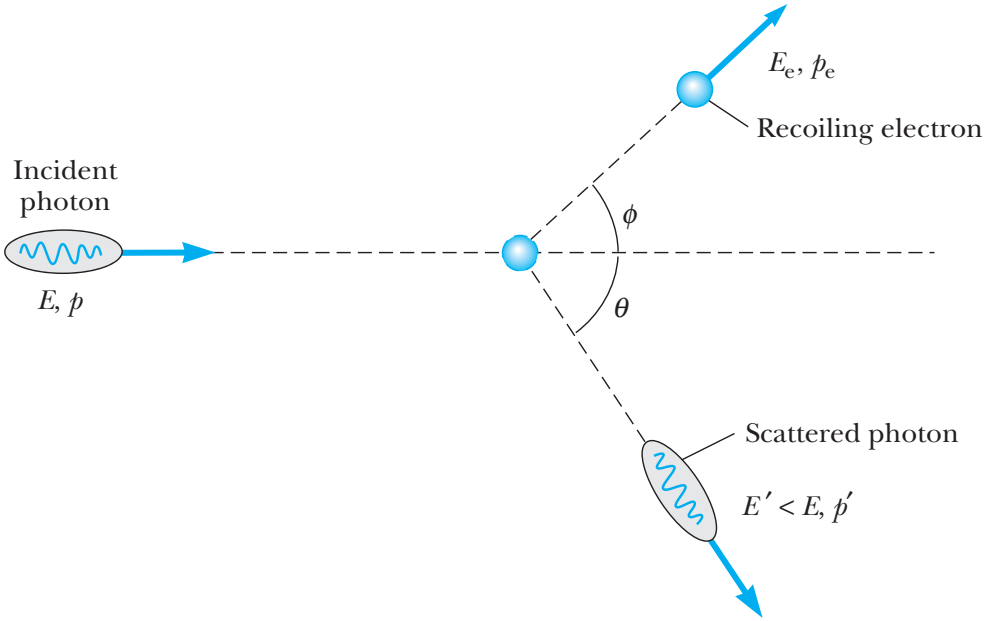


(a)



(b)

Diagram representing Compton scattering of a photon by an electron.



Conservation of energy

$$E + m_e c^2 = E' + E_e$$

Conservation of momentum

$$p = p' \cos\theta + p_e \cos\phi$$

$$p' \sin\theta = p_e \sin\phi$$

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Therefore

$$p_e^2 = (p')^2 + p^2 - 2pp' \cos\theta$$

With De Broglie relation

$$p_{\text{photon}} = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda}$$

We have

$$E_e = hf - hf' + m_e c^2 \qquad E_e^2 = p_e^2 c^2 + m_e^2 c^4$$

Finally

$$p_e^2 = \left(\frac{hf'}{c}\right)^2 + \left(\frac{hf}{c}\right)^2 - \frac{2h^2 ff'}{c^2}\cos\theta$$

$$\lambda' - \lambda_0 = \boxed{\frac{h}{m_e c}}(1 - \cos\theta)$$

→ **Compton wavelength**

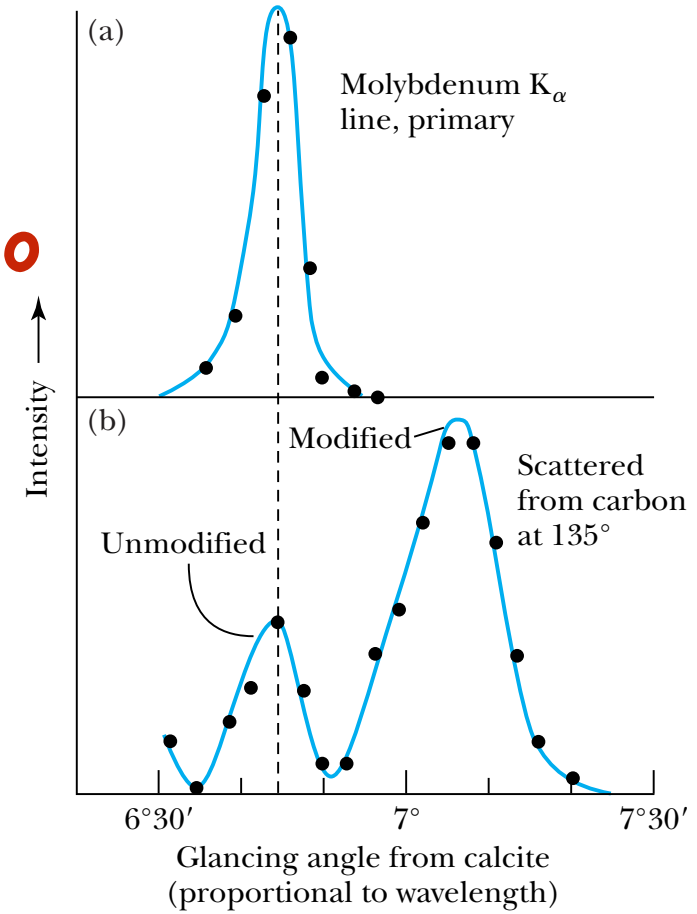
$\lambda_C = 2.426 \times 10^{-3}$ nm

6.3 Compton scattering

Compton's original data showing

1.the primary x-ray beam from Mo unscattered

2. the scattered spectrum from carbon at 135° showing both the modified and unmodified wave.



Pair Production

In the photoelectric effect, bremsstrahlung, and the Compton effect, we have studied exchanges of energy between photons and electrons. Have we covered all possible mechanisms?

For example, can the kinetic energy of a photon be converted into particle mass and vice versa?

It would appear that if none of the conservation laws are violated, then such a process should be possible.

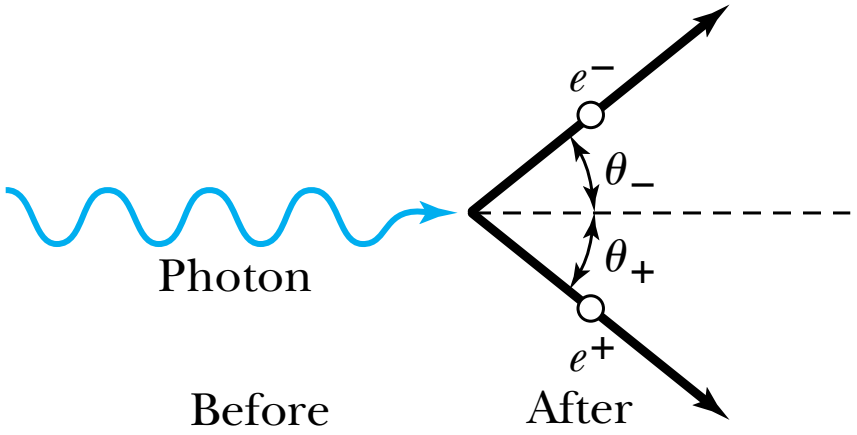
Pair Production

Experiments show that a photon's energy can be converted entirely into an electron and a positron in a process called pair production. The reaction is

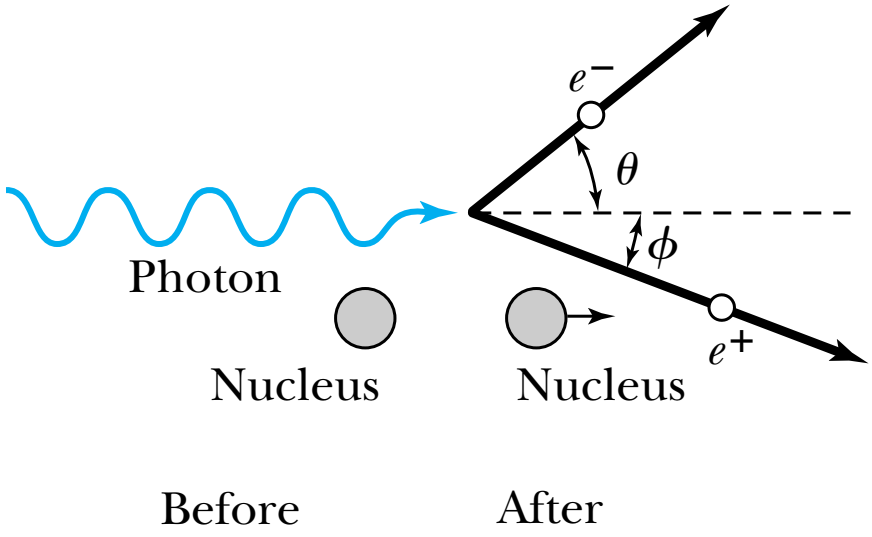
$$\gamma \rightarrow e^{+} + e^{-}$$

However, this process occurs only when the photon passes through matter, because energy and momentum would not be conserved if the reaction took place in isolation. The missing momentum must be supplied by interaction with a nearby massive object such as a nucleus.

Pair Production



(a) Free space (**cannot occur**)



(b) Beside nucleus

Pair Production

Consider the conversion of a photon into an electron and a positron that takes place inside an atom where the electric field of a nucleus is large. The nucleus recoils and takes away a negligible amount of energy but a considerable amount of momentum. The conservation of energy will now be

$$hf = E_+ + E_- + \text{K.E. (nucleus)}$$

The photon energy must be at least equal to $2m_ec^2$ in order to create the masses of the electron and positron.

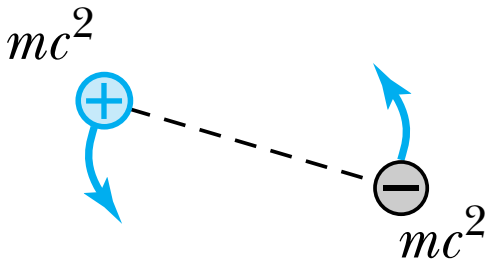
$$hf > 2m_e c^2 = 1.022\ \text{MeV} \qquad \text{(for pair production)}$$

Pair Annihilation

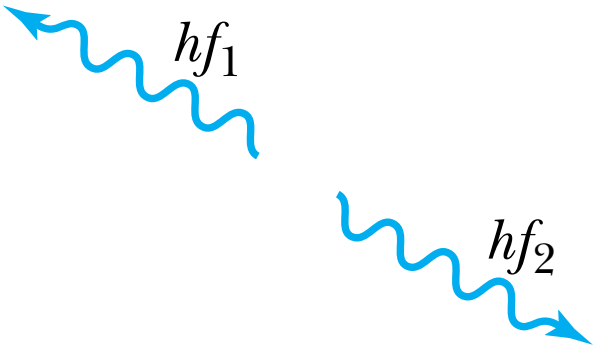
Eventually the electron and positron annihilate each other (typically in 10^{-10} s), producing electromagnetic radiation (photons). The process

$$e^+ + e^- \rightarrow \gamma + \gamma$$

is called pair annihilation.



(a) Positronium, before decay (schematic only)



(b) After annihilation

Pair Production

The conservation laws for the process will be

$$2m_e c^2 \approx hf_1 + hf_2$$

$$0 = \frac{hf_1}{c} - \frac{hf_2}{c}$$

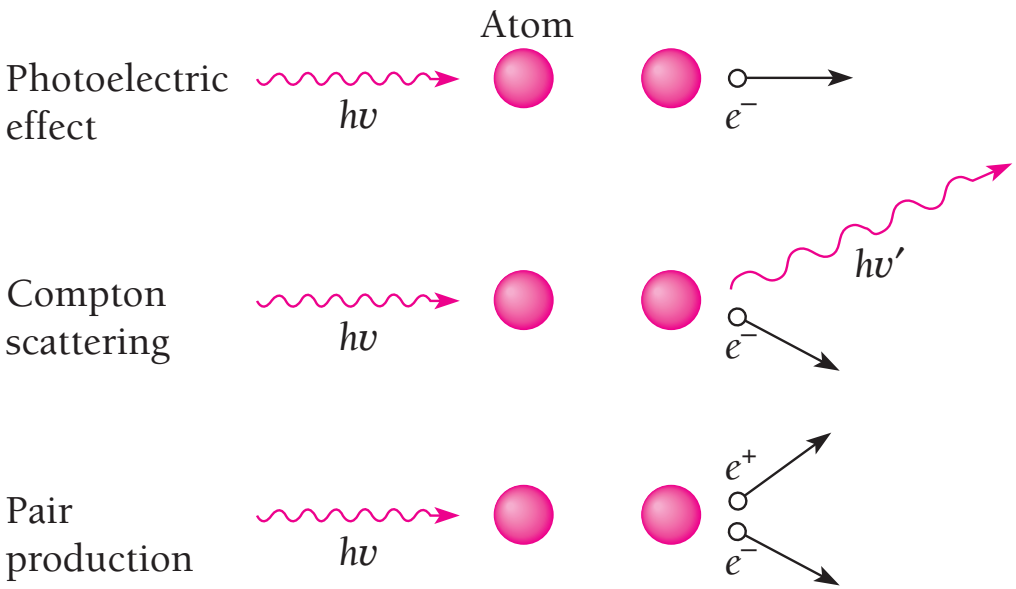
The frequencies are identical

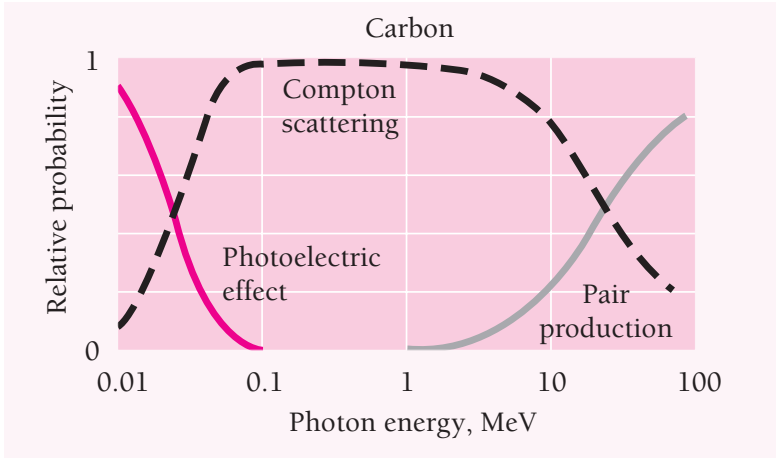
$$hf = m_e c^2 = 0.511 \text{ MeV}$$

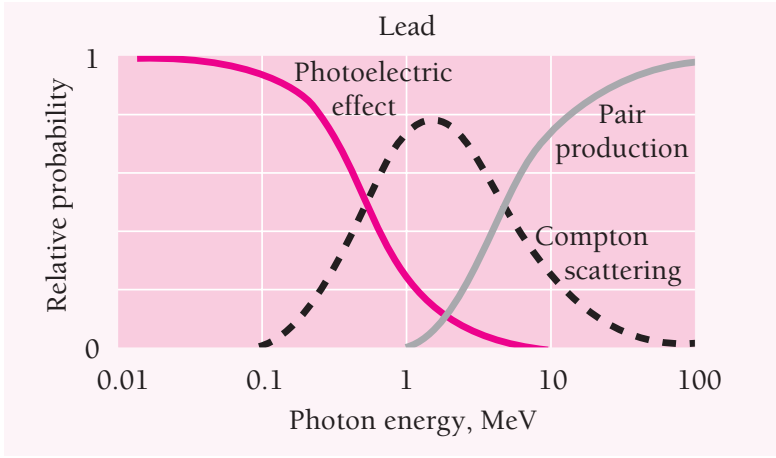
In other words, the two photons from positronium annihilation will move in opposite directions, each with energy 0.511 MeV. This is exactly what is observed experimentally.

6.4 Photon absorption

The three chief ways in which photons of light, x-rays, and gamma rays interact with matter







6.4 Photon absorption

The intensity I of an x- or gamma-ray beam is equal to the rate at which it transports energy per unit cross-sectional area of the beam.

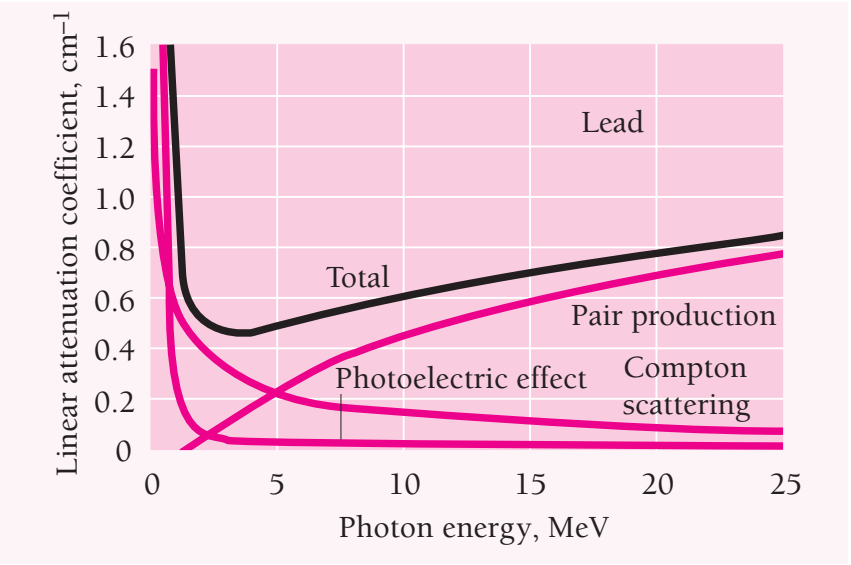
$$-\frac{dI}{I} = \mu\, dx$$

Radiation intensity I

$$I = I_0 e^{-\mu x}$$

Absorber thickness

$$x = \frac{\ln\,(I_0/I)}{\mu}$$



6.4 linear absorption coefficient

Linear Absorption Coefficient (cm^{-1})					
λ (pm)	**Air**	**Water**	**Aluminum**	**Copper**	**Lead**
10		0.16	0.43	3.2	43
20		0.18	0.76	13	55
30		0.29	1.3	38	158
40		0.44	3.0	87	350
50	8.6×10^{-4}	0.66	5.4	170	610
60	1.3×10^{-3}	1.0	9.2	286	1000
70	1.95×10^{-3}	1.5	14	430	1600
80	2.73×10^{-3}	2.1	20	625	
90	3.64×10^{-3}	2.8	30	875	
100	4.94×10^{-3}	3.8	41	1200	
150	1.56×10^{-2}	12	124		
200	3.64×10^{-2}	28	275		
250	6.63×10^{-2}	51	524		

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原子物理学

6-5, 6-8

The Physics of Atoms and Quanta

18.1, 18.2, 18.4, 18.8

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The Physics of Atoms and Quanta (H. Haken, H. C. Wolf)

18.1 What is the shortest possible wavelength for bremsstrahlung observed when an electron which has been accelerated through a potential difference of 40 kV is stopped by the anticathode of an x-ray tube? In what region of the electromagnetic spectrum does this wavelength lie?

The Physics of Atoms and Quanta (H. Haken, H. C. Wolf)

18.2 The K_α line of cobalt is at 1.785 Å. What is the energy difference between the 1s and 2p orbitals in cobalt? Compare this result with the energy difference between the 1s and 2p orbitals in hydrogen (that is, the first Lyman line). Why is the difference much greater for cobalt than for hydrogen?

18.4 The maximum energy of the characteristic x-rays emitted by a sample of unknown composition corresponds to the wavelength 2.16 Å. Of what element does the sample consist?

The Physics of Atoms and Quanta (H. Haken, H. C. Wolf)

18.8 a) The K absorption edge of tungsten is at 0.178 Å, and the wavelengths of the lines of the K series are (ignoring the fine structure) $K_\alpha = 0.210$ Å, $K_\beta = 0.184$ Å, and $K_\gamma = 0.179$ Å. Sketch the energy levels of W and give the energies of the K, L, M and N shells.

b) What is the minimum energy required to excite the L series in tungsten? What is the wavelength of the L line?